How well can AI agents reproduce research paper methods?

30 papers across different data science topics, each with an implementation on GitHub.

OUR TASK

Evaluate how well an AI agent can reproduce the main methods of a paper, especially where it fails and where humen intervention is needed.

01

Build the pipeline

Turn a paper into code, run it, and compare the results with the authors' implementation.

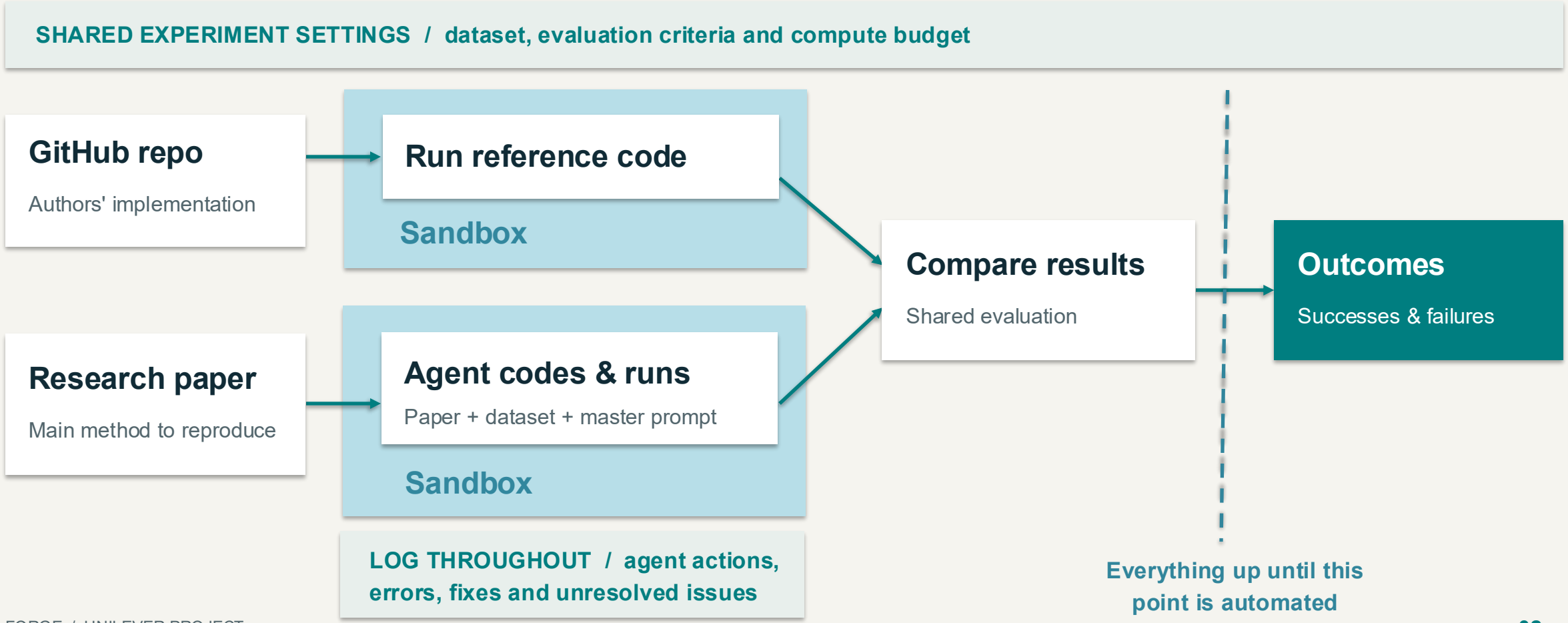
02

Analyze the 30-paper study

Identify success and failure patterns across paper types and produce a summary report.

From paper to code to comparison

A short view of the process, repeated for each paper.



From a working pipeline to a 30-paper study

Task	Proposed lead(s)	Deliverable
Build the pipeline	Arut + Adelaide	Working paper-to-code workflow
Run experiments across 15 papers (half)	All four teammates	Results table + pain points
Improve pipeline (if needed)	Anna + Owen	Improved paper-to-code workflow
Run experiments across 30 papers (all)	All four teammates	Results table + implementation logs
Qualitative evaluation per paper	All four teammates	Code quality + method fidelity assessments
Analyze success / failure patterns	All four teammates	Patterns across paper types
Write the summary report	All four teammates	Findings

Every two weeks: present a concrete deliverable to the Unilever project lead.